

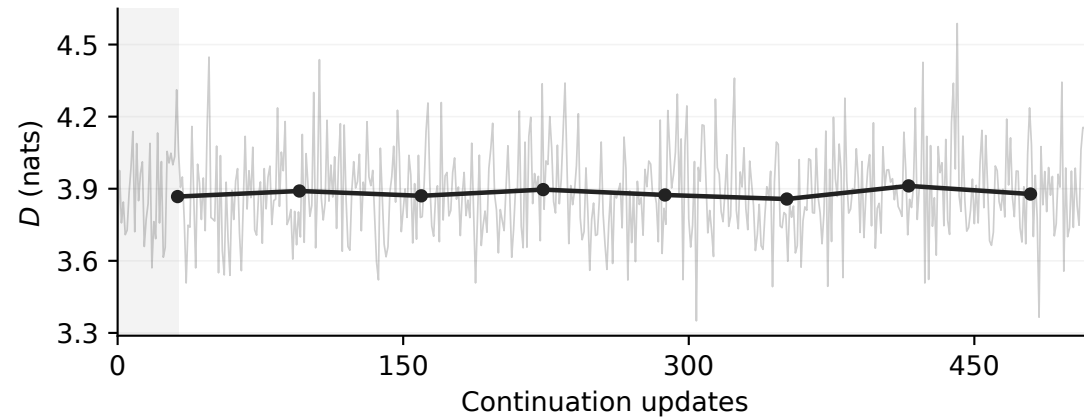
SGD (smaller rate)
 effective $h=3e-08$

Adam (calibrated)
 raw $\eta=3.03e-09$

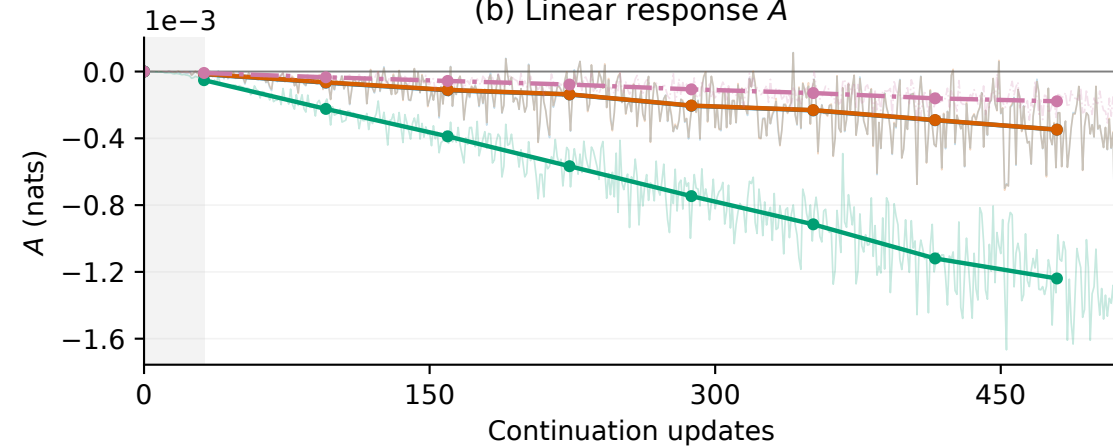
Head only
 raw $\eta=3e-08$

Heavy Ball (smaller rate)
 effective $h=3e-08$

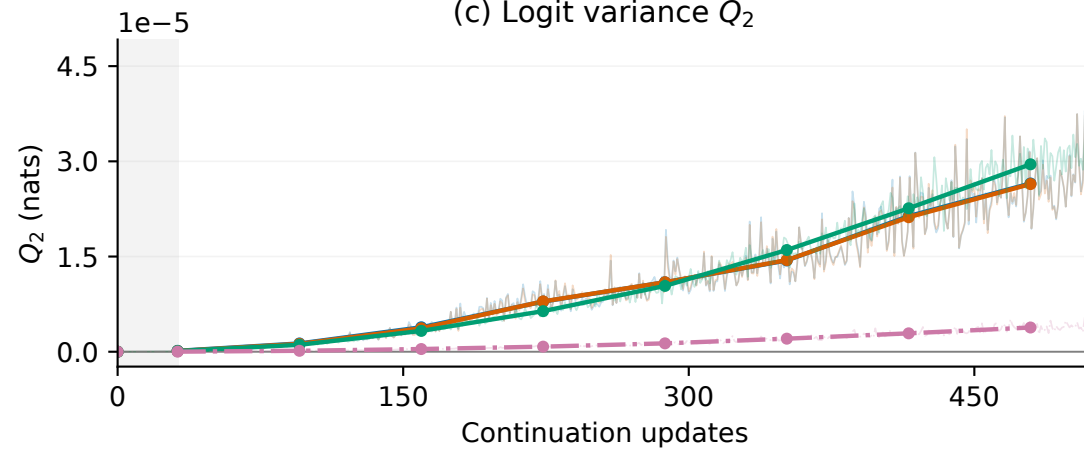
(a) Shared frozen difficulty D



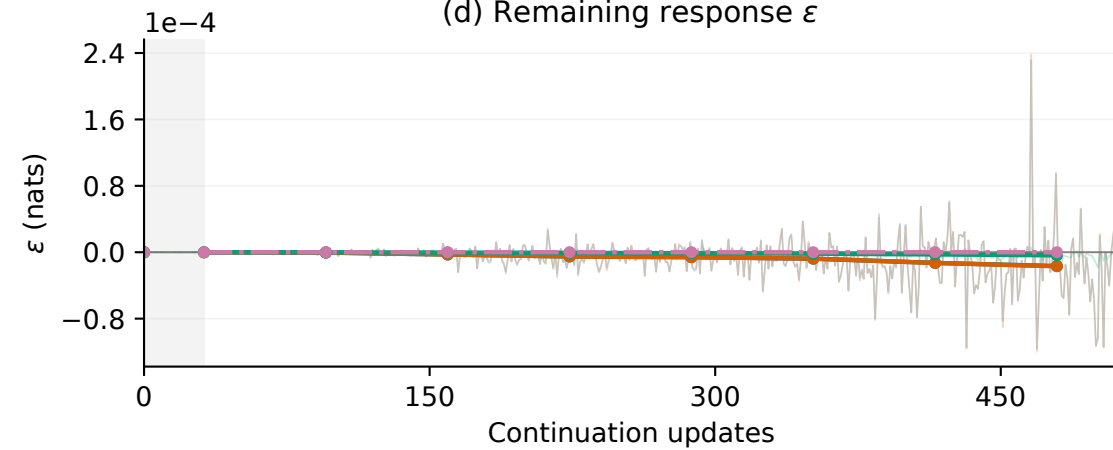
(b) Linear response A



(c) Logit variance Q_2



(d) Remaining response ε



Light lines: all observed arriving batches. Solid/dashed lines with dots: nonoverlapping 64-step means.
 Terminal local-scale protocol (adaptive); permutation 1 of 3. Smaller-rate/calibrated settings only; larger-rate results are in the full figure.
 Heavy Ball raw $\eta = 0.1h$; Adam rate matches one reset-step Q_2 , not its whole trajectory.
 Grey: 32-update warmup. Shared batch order; separate vertical scales; all nats. Rate labels do not imply stability regimes.